Project 12 / Phase 1

Smart Waste Segregation

Handcrafted features and calibrated classical classification

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A material suggestion from a photograph

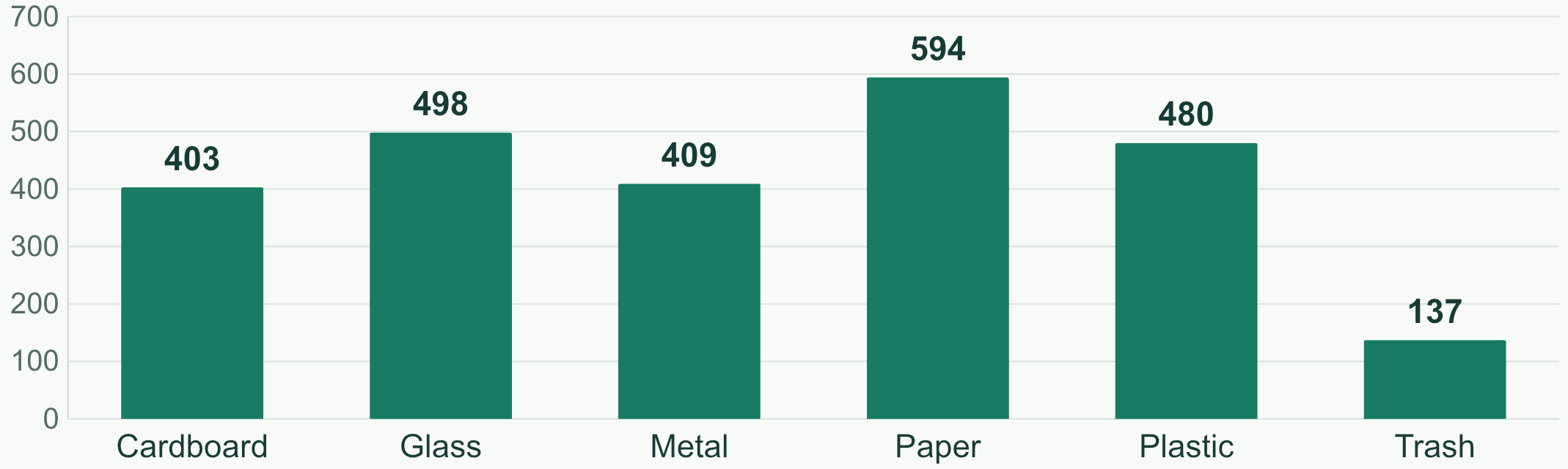
Cardboard Glass Metal
Paper Plastic Trash

Single-photo input with six known categories

Uncertainty response when confidence is below 0.52

Local recycling rules and contamination remain outside the classifier’s scope.

2,521 usable images after the audit



2,527 source images − 6 conflicting-label duplicates = 2,521 included images

Training, threshold selection and test stay separate

Partition	Images	Role
Training	1,764	Model selection and calibration
Validation	380	Abstention threshold only
Test	377	Frozen final evaluation

Five saved training folds with seed 42

Detected duplicate groups never cross partitions

Baseline features capture color, shape and texture

Descriptor	What it represents	Dimensions
HSV histograms	Color distribution	32
HOG	Local edge orientations	1,764
Uniform LBP	Local texture patterns	10
Combined vector	Fixed order, float32	1,806

Shared preprocessing: RGB, preserved aspect ratio, white padding, 128 × 128 canvas.

Spatial texture adds detail to the representation

6 regions × 460 values = 2,760 inputs

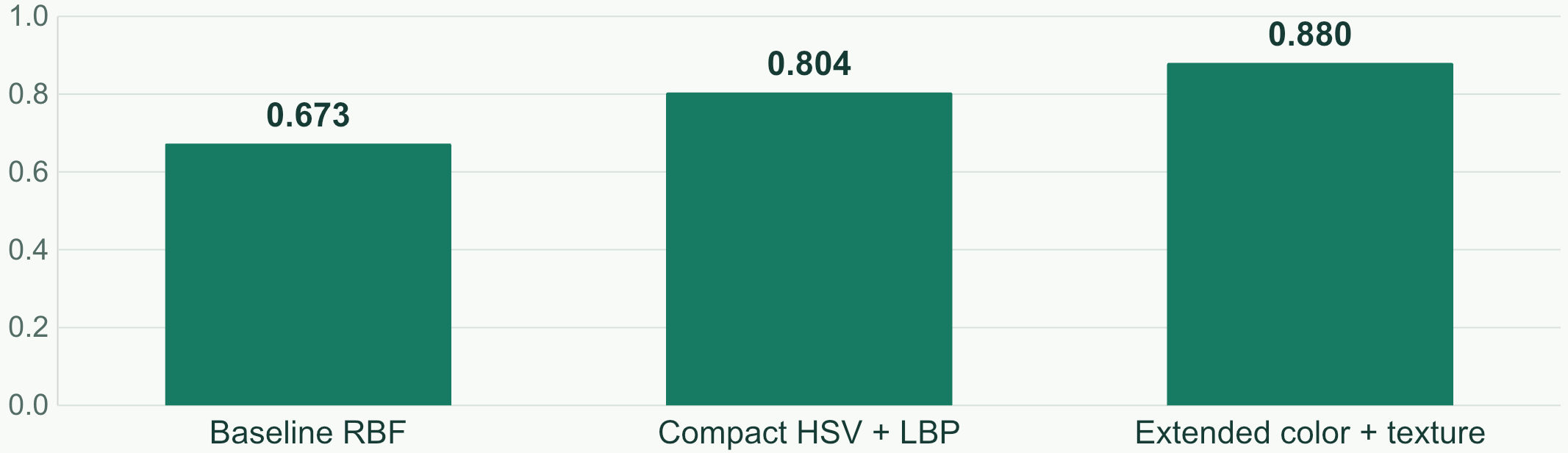
Color: HSV, RGB, Lab and joint hue–saturation

Texture: multi-scale uniform and rotation-sensitive LBP

Winner: 576 color + 1,032 texture = 1,608 selected features

The extended representation uses color and texture; HOG remains in the baseline study.

The extended SVM reaches 0.8803 mean CV F1



Square root + per-image L2 normalization

RBF SVM: C = 10, γ = 10, no class weighting

Random Forest improves with richer descriptors



C = color, T = texture. Same 24 configurations per family, 480 fold fits overall.

More features do not always improve the model

Comparison	Smaller representation	Larger representation
Initial Random Forest	32 HSV: 0.7366	1,806 combined: 0.7090
Extended RBF SVM	1,608 selected: 0.8803	2,760 full: 0.8696

We selected the extended RBF SVM using mean training-fold macro F1.

Calibration and the abstention rule

Accept when max probability ≥ 0.52

Five base models, each paired with a held-out-fold sigmoid calibrator

Validation coverage: 373 / 380 = 98.16%

Validation accepted accuracy: 336 / 373 = 90.08%

Threshold selection uses validation only. Test predictions never tune the threshold.

Frozen test result: 0.8999 macro F1

Metric	All test images	Accepted at 0.52
Images	377	371
Accuracy	90.19%	91.37%
Macro F1	0.8999	0.9129
Coverage	100%	98.41%

6 abstentions: 5 incorrect predictions and 1 correct prediction withheld

Errors concentrate in visually similar materials

True / Pred.	Cardbd.	Glass	Metal	Paper	Plastic	Trash
Cardboard	**57**	0	0	1	2	0
Glass	0	**68**	3	3	1	0
Metal	1	6	**54**	0	0	0
Paper	1	2	1	**81**	1	3
Plastic	1	4	3	2	**61**	1
Trash	1	0	0	0	0	**19**

84.72%

Plastic recall
61 of 72 correct

20

Trash test images

Counts from the saved 377-image test evaluation.

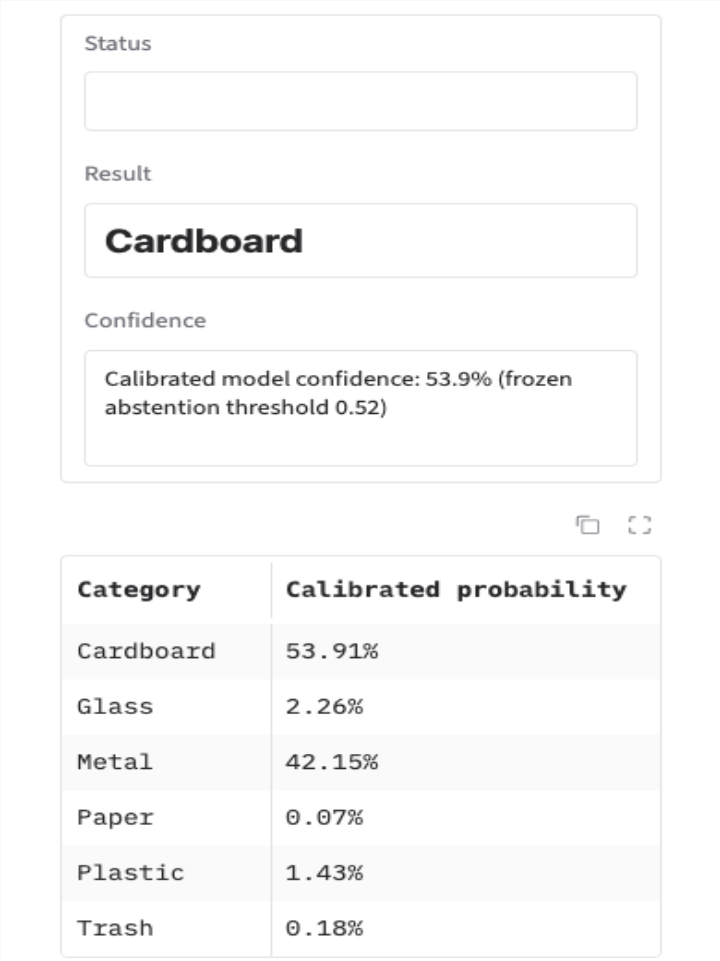
Photo upload demo

Upload one image

View the class and six probabilities

Inspect the uncertainty response

Clear the input and prediction



Screenshot: generated patch, real evaluated model. Functional illustration only.

What this result does and does not establish

Controlled TrashNet photos differ from everyday scenes

Unknown categories and mixed materials remain unverified

Calibration error remains: ECE = 0.1325

Historical evaluation source has a reconstruction limitation

Phase 2: compare learned representations and test lighting, background and viewpoint changes.

Six contributions, one reproducible pipeline

Member	Primary work
Om Yadav	Data verification and preprocessing
Aditya Bharadwaj	Descriptors and feature caching
Shourya Bafna	Training protocol, SVM and integration review
Daksh Batra	Random Forest studies
Prajjwal Tripathi	Calibration and frozen evaluation
Saksham Mighlani	Demo and submission integration

Questions

Backup: the selected SVM pipeline

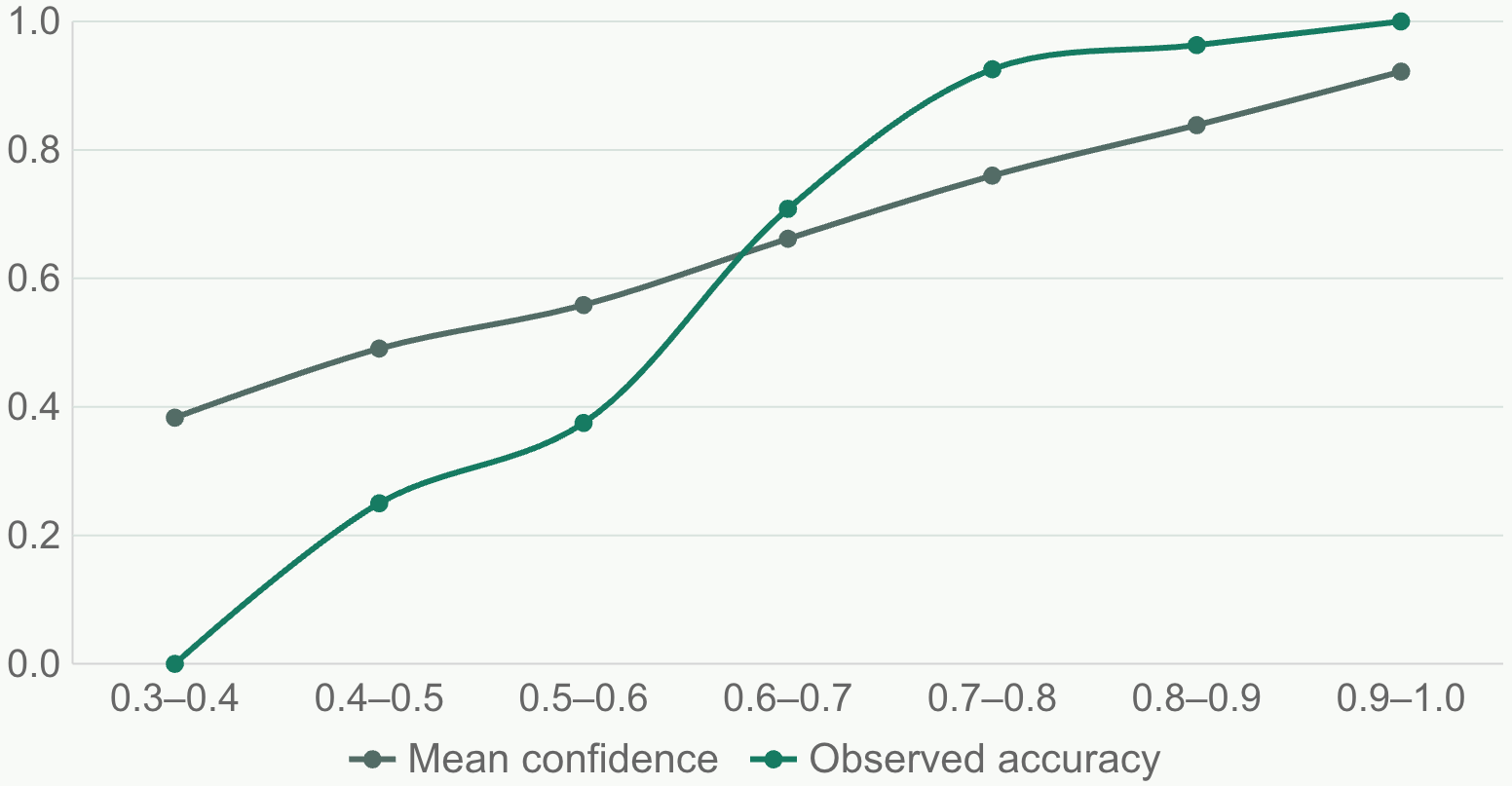
2,760 inputs with 1,608 columns selected inside the pipeline

Elementwise square root, then normalization to unit L2 norm

$K(x, z) = \exp(-\gamma \|x - z\|^2)$

$C = 10$, $\gamma = 10$, class weight = none

Backup: calibration is still imperfect



0.1325

ECE, 10 bins

0.4599

Log loss

0.1828

Brier score

Backup: references and experiment records

TrashNet: Thung and Yang (2016)

Descriptors: Dalal and Triggs (2005), Ojala et al. (2002)

Models: Cortes and Vapnik (1995), Breiman (2001)

Confidence: Platt (1999), Niculescu-Mizil and Caruana (2005)

Selective classification: El-Yaniv and Wiener (2010)

Repository: github.com/shourya2006/smart-waste-segregation